

UNIT 4: Force, Work, Energy, and Power

Subject: Physics

Grade: Grade 9

Force, Newton's laws, friction, work, energy, and power with core formulas.

The Concept of Force

A force is a push or pull that can change an object's state of motion or deform it. Since force has magnitude and direction, it is a vector quantity.

Unit of force: newton (N), where $1\text{ N} = 1\text{ kg}\cdot\text{m}/\text{s}^2$.

Contact and Non-Contact Forces

Some forces act by touching (contact), while others act at a distance (non-contact).

Type	Examples
Contact forces	Applied force, normal force, friction, spring force
Non-contact forces	Gravitational force, magnetic force, electric force

Newton's Laws of Motion

Newton's laws explain rest, motion, and acceleration relationships.

Law	Statement	Key Idea
First Law	A body remains at rest or in uniform straight-line motion unless acted on by an unbalanced force.	Inertia
Second Law	Acceleration is proportional to net force and inversely proportional to mass.	$F = ma$
Third Law	Every action has an equal and opposite reaction.	$F_{12} = -F_{21}$

Weight, Friction, and Motion

Weight is gravitational force: $W = mg$. Mass is constant, but weight changes with local gravity.

Friction opposes motion and depends on surface type and normal force.

- static friction: before sliding starts
- kinetic friction: during sliding/rolling

Work, Energy, and Power

Work is done when a force causes displacement in the direction of force: $W = F_{\text{parallel}} \times S$.

Energy is the ability to do work.

- Kinetic energy: $E_k = \frac{1}{2} mv^2$
- Gravitational potential energy: $E_p = mgh$
- Mechanical energy: $E_m = E_k + E_p$

Power is rate of doing work: $P = W/t = E/t$. Unit: watt (W), where $1 \text{ W} = 1 \text{ J/s}$.

Worked Examples for Unit 4

Practice tasks:

- A 10 N force acts on a 2 kg block. Find acceleration.
- A 100 N force causes acceleration 2 m/s^2 . Find mass.
- Find weight of a 10 kg body on Earth.
- A 60 N force moves a box 12 m (parallel force). Find work done.
- Find vertical force needed to lift a load 3 m if work is 600 J.
- A 0.2 kg ball moves at 20 m/s. Find kinetic energy.
- Find speed of a 1 kg ball with kinetic energy 50 J.
- Find potential energy of 30 kg at height 50 m ($g = 10 \text{ m/s}^2$).
- A machine lifts 100 kg through 5 m in 2 s. Find power.
- A crane does $1.5 \times 10^5 \text{ J}$ in 10 s. Find power.